

BSV Prediction Market — Execution Roadmap

Practical execution plan for launching a Polymarket-style binary prediction market with native satoshi settlement.

This document establishes the exact engineering scope, architectural split, LMSR liquidity mechanics, realistic VPS server budget, team composition, and intensive 4-week fast-track execution timeline required to launch a production-grade, self-hosted binary prediction market on the BSV blockchain.



MVP LAUNCH TIME

4 Weeks

4 rapid weekly sprint phases

REQUIRED TEAM

Internal BSVA Team

5 roles (1 lead + 2 full-stack + 2 specialists)

DEV INFRASTRUCTURE COST

\$1,200 USD

\$0 labor (internal staff) + Staging VPS & API tooling

STARTING LIQUIDITY

10 BSV

Testing bootstrap pool, scaling dynamically to 1,000 BSV

1. EXECUTIVE SUMMARY

Launch a fast, intuitive binary prediction market powered by native BSV satoshis inside 4 weeks (1-month rapid sprint), executed entirely by internal BSV Association (BSVA) developers with zero external hiring overhead and reliable dedicated VPS server infrastructure.

- The Core Objective:** Build a clean, streamlined web platform where users connect their BSV wallet, deposit satoshis into an internal double-entry ledger, and trade binary YES/NO contracts on real-world events.
- The Architectural Strategy:** Keep deposits, withdrawals, and treasury sweeps strictly on-chain using native BSV transactions. Execute all order matching, price discovery, LMSR calculations, and balance updates off-chain in sub-millisecond memory structures to eliminate gas fees, friction, and confirmation delays during active trading.
- External Data Separation:** Import all market questions, rules, opening/closing schedules, and final resolution outcomes from Kalshi via a clean provider adapter. Calculate and govern all YES/NO prices strictly internally using an integer-safe Logarithmic Market Scoring Rule (LMSR) engine.
- Financial & Resource Viability:** Complete full production delivery using internal BSVA team members (\$0 external labor budget), \$1,200 in staging VPS servers and API tooling during the 4-week build, \$1,700 in monthly production VPS operating costs, and an initial 10 BSV (1,000,000,000 satoshis / \$140 USD at \$14/BSV) bootstrap testing pool that scales dynamically up to 100 BSV and 1,000 BSV as user adoption grows.

2. THE MVP IN ONE VIEW

A focused, high-throughput binary market engineered for immediate user utility and exact financial accounting.

Product Specifications

CORE SCOPE

- Contract Type:** Binary YES / NO outcomes only.
- Base Currency:** Native BSV Satoshis (1 BSV = 100,000,000 sats).
- Catalogue Size:** 10 to 20 curated simultaneous public markets.
- External Provider:** Kalshi (via modular REST/Webhook adapter).
- Price Engine:** Internal Off-chain LMSR Market Maker.
- Trading Operations:** Instant buy and sell quotes before market close.
- Settlement Mechanics:** Automatic balance credit immediately upon confirmed result.

Operational & Risk Parameters

BOOTSTRAP PHASE 1

- Contract Payout Unit:** 100,000 satoshis (\$0.014 per contract at \$14/BSV assumption).
- Trading Fee:** 1.0% flat fee charged on trade volume, reserved in fee pool.
- Minimum Order:** 1,000 satoshis (0.01 contracts).
- Maximum Order (Phase 1):** 1,000,000 satoshis (10 contracts per single trade).
- Position Limit (Phase 1):** 5,000,000 satoshis max net exposure per user per market.
- LMSR Parameter ('b' Phase 1):** 200,000 satoshis per market (2 contracts depth).
- Max Loss / Market (Phase 1):** 138,630 satoshis exact integer ceiling per market.

3. HOW THE PRODUCT WORKS

The platform operates as an automated counterparty for binary real-world predictions. Every market poses a clear YES or NO question with a definitive resolution date. Users deposit native BSV into their platform account and buy contract shares representing either YES or NO. When users buy YES shares, the internal LMSR formula automatically increases the YES price and decreases the NO price in direct proportion to order size. The sum of the implied YES price and NO price always equals approximately 100,000

satoshis (one contract unit). Users can sell their position back to the automated market maker at any moment before the closing timestamp to take profits or cut losses. Once the market closes and the external provider confirms the final outcome, winning contract shares automatically redeem for exactly 100,000 satoshis each, while losing shares expire at zero. All balances are updated instantly, and users withdraw their satoshis directly to their personal BSV wallet.

DIAGRAM 3A — VISUAL SCHEMA: HOW BINARY CONTRACTS WORK & HOW THE POOL DEFENDS AGAINST ABUSE

A visual explainer designed so that anyone—even with zero technical or financial background—can instantly understand what owning a prediction contract means and how the liquidity pool is mathematically defended against manipulation.

PART 1: WHAT DOES OWNING A PREDICTION CONTRACT MEAN? (STEP-BY-STEP IN SATOSHIS)

STEP 1 • THE STANDARD

1 Contract = 100,000 Satoshis

Every market poses a clear binary question (e.g., "Will Bitcoin surpass \$150,000 by Dec 31?"). Every single contract unit has a fixed payout value of exactly **100,000 satoshis** (\$0.014 at \$14/BSV). If YES happens, YES pays 100,000 sats. If NO happens, YES pays 0.

Winning Payout = 100,000 Sats

STEP 2 • PURCHASING

How You Buy & What It Costs

If the market currently estimates a 60% probability for YES, the price of 1 YES contract is **60,000 satoshis**. You pay 60,000 sats to buy 1 share. You now own digital claim rights: if YES occurs, the platform will give you 100,000 sats (a **40,000 sat net profit**).

Buy YES at 60,000 Sats

STEP 3 • TRADING EARLY

Selling Before Market Close

You do not have to wait until the end of the year! If positive news arrives tomorrow and demand drives the YES price up to **85,000 satoshis**, you can sell your share back to the platform instantly for 85,000 sats, locking in a **25,000 sat profit** early.

Sell Early for 85,000 Sats

STEP 4 • RESOLUTION

Automatic Final Settlement

When the market reaches its closing timestamp, Kalshi confirms the real-world outcome. If YES is confirmed, the internal ledger automatically converts every held YES share into exactly **100,000 satoshis** inside your withdrawable balance.

Redeem for 100,000 Sats

PART 2: HOW DOES THE LIQUIDITY POOL DEFEND AGAINST ABUSE AND EXAGGERATED PAYOUTS?

Why can't an attacker or whale drain the platform's bankroll? The platform uses four concrete mathematical and operational shields:

Shield 1: Dynamic Price Scaling (LMSR Curve)

MATHEMATICAL
ARMOR

You cannot buy unlimited YES shares at a cheap price. The LMSR formula acts like an automated scale: every share purchased instantly increases the price of the next share towards 100,000 satoshis. If a whale attempts to buy huge volume, the price climbs to **95,000, 98,000, and 99,900 sats**, making it mathematically impossible to extract more payout than the exact formula permits.

Small Trade: Price rises slightly → Whale Trade: Price approaches 100,000 sats

Shield 2: Capped Worst-Case Loss ($b \cdot \ln(2)$)

BOUNDED
RISK

Before any market opens, the platform calculates its absolute worst-case mathematical loss: exactly **138,630 satoshis** (\$0.019) per market when $b = 200,000$. Even if 10,000 traders all bet on YES and YES wins, the platform bankroll loss **can never exceed 138,630 satoshis** because the winning payouts are funded primarily by the satoshis deposited by the losing NO bettors!

Max Loss Locked Before Open: Exactly 138,630 Satoshis Ceiling

Shield 3: Hard Order & Exposure Caps

ANTI-SPAM / ANTI-WHALE

The platform enforces strict hard caps at the API level: a maximum single order size of **1,000,000 satoshis** (10 contracts) and a maximum net exposure cap of **5,000,000 satoshis** per user per market. This prevents automated bots or rogue scripts from flooding the order book or monopolizing market liquidity during the bootstrap testing phase.

Max Trade: 1M Sats | Max User Position: 5M Sats

Shield 4: 100% Customer Fund Separation

SEGREGATED
LEDGER

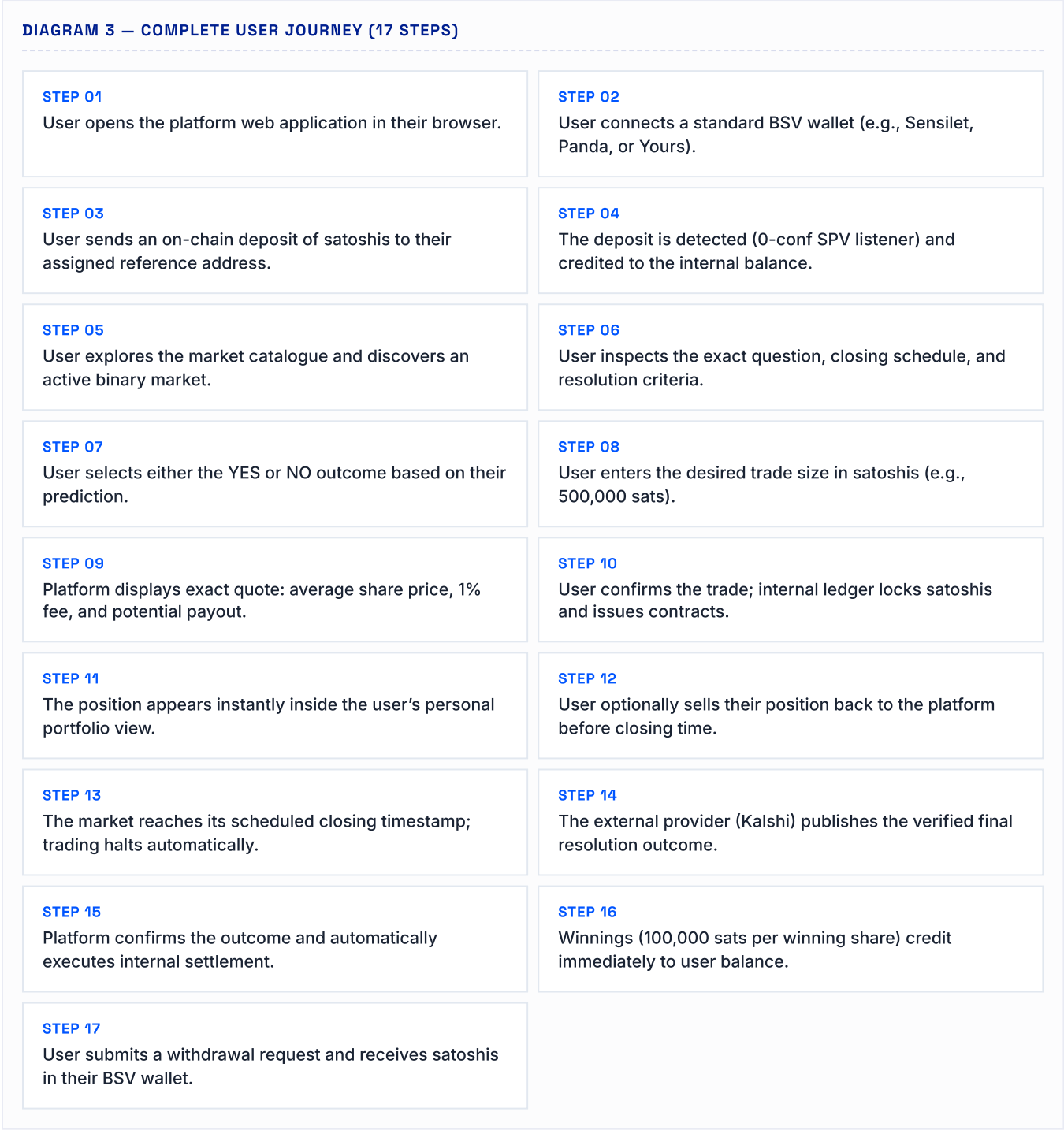
Customer deposits are locked in a segregated **Withdrawal Reserve** box inside the double-entry accounting ledger. The automated market maker **ONLY** trades using the dedicated **Market Liquidity Reserve** (10 BSV bootstrap bankroll). A user can never extract or abuse another user's deposit principal under any circumstances.

Customer Deposits: 100% Backed & Segregated



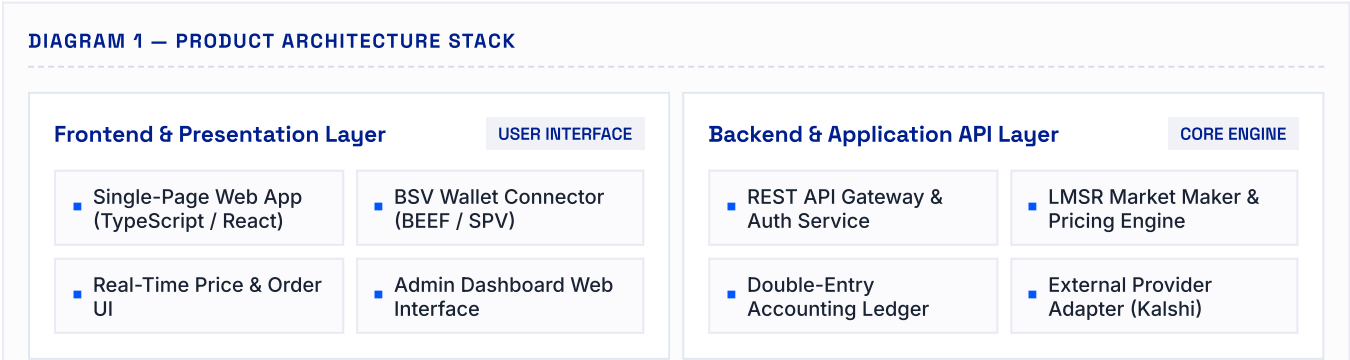
4. COMPLETE USER JOURNEY

A seamless 17-step flow hiding all complex cryptographic terminology from the end user.



5. PRODUCT ARCHITECTURE

Modular system architecture isolating external data feeds, core trading mechanics, and blockchain indexing on reliable dedicated VPS infrastructure.



Data Persistence & Processing Layer

STATE & WORKERS

■ PostgreSQL (Users, Trades, Ledger DB)

■ Redis In-Memory State & Lock Cache

■ Result & Lifecycle Worker Service

■ Deposit Detection & SPV Worker Service

Blockchain & Custody Layer

BSV NETWORK

■ Deposit Hot Wallet & Address Pool

■ Treasury & Liquidity Bankroll Wallet

■ Withdrawal Processing & Signing Service

■ BSV Network SPV Verifier & Worker

6. ON-CHAIN VS. OFF-CHAIN RESPONSIBILITIES

Why this architecture is mandatory: Executing every micro-trade directly on-chain requires user wallet signatures and broadcast overhead for every single order, which slows UX and complicates order matching. By restricting on-chain operations exclusively to fund transfers (deposits and withdrawals) while handling order matching and accounting inside an off-chain double-entry ledger, the platform delivers **instant pricing (0ms latency)**, **zero gas fees per trade**, **simple upgrades**, and an **effortless user experience**.

DIAGRAM 2 — ON-CHAIN VS. OFF-CHAIN SEPARATION MATRIX

On-Chain Scope

BSV BLOCKCHAIN ONLY

- **User Deposits:** Incoming satoshi transactions sent from user wallets to assigned platform addresses.
- **User Withdrawals:** Outgoing satoshi transactions broadcast from platform hot wallet to user BSV addresses.
- **Treasury Transfers:** Internal rebalancing between operating hot wallet, liquidity reserve, and cold storage.
- **Optional Settlement Anchor:** Writing immutable final market settlement receipts and outcome hashes to OP_RETURN.

Off-Chain Scope

INTERNAL HIGH-SPEED LEDGER

- **User Balances:** Real-time internal satoshi credit/debit balances.
- **Double-Entry Ledger:** Exact debit/credit accounting records preventing accidental creation or destruction of funds.
- **Trades & Positions:** Execution records, YES/NO contract ownership, and portfolio tracking.
- **LMSR Pricing & Liquidity Engine:** Dynamic price calculations, spread control, and exposure calculations.
- **Market Data & Result Processing:** Importing Kalshi metadata, scheduling, and outcome processing.
- **Settlement & Admin Operations:** Winning contract payouts, fee sweeps, pause controls, and refunds.

7. MARKET LIFECYCLE & RESULT FLOW

Every market transitions through a deterministic sequence of states governed by automated workers and admin checkpoints.

DIAGRAM 5 — EXACT MARKET LIFECYCLE & EXCEPTIONAL FLOWS

IMPORTED

→

REVIEWED

→

OPEN

→

TRADING

→

CLOSED

→

AWAITING RESULT

→

RESULT RECEIVED

→

CONFIRMED

→

SETTLED

→

WITHDRAWABLE

EXCEPTIONAL OVERRIDES:

PAUSED

VOIDED

REFUNDED

Standard Lifecycle Stages

- **Imported:** Worker pulls new binary market question, rules, and schedules from Kalshi API adapter.
- **Reviewed:** Admin verifies clarity, confirms resolution criteria, and assigns initial LMSR parameter 'b'.
- **Open:** Market becomes visible in catalogue; required max loss satoshis ('138,630 sats') lock in reserve.
- **Trading:** Users actively buy and sell YES/NO contracts; LMSR adjusts quotes dynamically based on volume.
- **Closed:** Scheduled closing timestamp arrives; order acceptance immediately halts; positions freeze.
- **Awaiting Result:** Market waits for the external provider to finalize and publish the real-world resolution.
- **Result Received:** Kalshi adapter detects definitive YES or NO outcome and queues settlement payload.
- **Confirmed:** Admin operator (or automated 2-of-2 check) validates external outcome against official rules.
- **Settled:** Ledger credits 100,000 satoshis per winning contract to user balances and zeroes losing shares.
- **Withdrawable:** Winning users request immediate satoshi withdrawal to their personal BSV wallet.

Exceptional Operational Overrides

- **Paused:** Admin halts trading instantly due to breaking news, feed anomalies, or exposure threshold breaches. Users cannot buy or sell while paused, but existing positions remain intact.
- **Voided:** Admin cancels a market if the underlying question becomes unresolvable, ambiguous, or if Kalshi voids the reference contract.
- **Refunded:** Following a voided market, the double-entry ledger automatically reverses all executed trades, returning exact initial satoshi principal and fees to every participating user balance.

8. LMSR MECHANICS & INTEGER-SAFE SATOSHI CALCULATIONS

The Logarithmic Market Scoring Rule (LMSR) acts as an automated market maker that always offers a price to buy or sell. The liquidity parameter 'b' controls how sensitive prices are to trade volume. A higher 'b' means larger trades move the price less, but increases the maximum potential loss the platform must reserve. To prevent floating-point rounding errors and satoshi discrepancies, all prices and balances are calculated and stored using exact 64-bit integer arithmetic ('BIGINT' in satoshis), scaling formulas by '10^8' during intermediate computation.

DIAGRAM 4 — MONEY AND POSITION FLOW & INTEGER-SAFE PARAMETERS (\$14/BSV ASSUMPTION)

Recommended LMSR Numerical Parameters (Bootstrap Phase 1)

Contract Payout Unit	100,000 satoshis (\$0.014 at \$14/BSV)
Initial YES / NO Price	50,000 satoshis (50.00%)
Minimum Trade Size	1,000 satoshis (0.01 contracts)
Maximum Trade Size (Phase 1)	1,000,000 satoshis (10 contracts)
Platform Trading Fee	1.00% flat on trade volume
LMSR Liquidity Parameter ('b' Phase 1)	200,000 satoshis (2 contracts depth)
Max Platform Loss / Market ('b * ln(2)')	138,630 satoshis exact ceiling (\$0.019)
Simultaneous Active Phase 1 Markets	10 active markets (1,386,300 sats reserved)

Exact Satoshi Numerical Example

Assume a new market opens with 'b = 200,000 satoshis'.

- 1. Initial State:** YES and NO cost exactly '50,000 sats' per share ('50%'). Platform reserves '138,630 sats' ('200,000 * 0.693147') from its bankroll.
- 2. User Trade:** A user deposits '101,000 sats' and buys YES shares. They pay '100,000 sats' principal + '1,000 sats' fee ('1%').
- 3. Price Adjustment:** With '100,000 sats' ('1 contract unit') added to YES demand, the integer LMSR engine updates the price of the next YES contract to '62,246 sats' ('62.25%') and NO drops to '37,754 sats' ('37.75%').
- 4. Resolution:** If YES wins, the user receives '100,000 sats' for every contract share owned, paid directly from the combined pool of user trade receipts plus the reserved '138,630 sat' bankroll buffer.

9. STARTING BANKROLL & POOL PROTECTION PLAN

Strict Fund Separation Rule: Customer deposits are never treated as platform revenue or market liquidity. The internal double-entry accounting engine enforces strict partitioning between Customer Funds, Market-Maker Capital, and Platform Revenue.

DIAGRAM 6 — LIQUIDITY POOL PARTITIONING & BANKROLL ALLOCATION (\$14/BSV ASSUMPTION)

CUSTOMER DEPOSITS Locked in Withdrawal Reserve. 100% backed at all times. Never used for trading subsidies or ops.		PLATFORM CAPITAL Allocated to Market Liquidity Reserve ($b * \ln(2)$ per open market) + Hot Wallet operating buffer.		
PLATFORM REVENUE Accumulates from 1% trading fees. Kept inside Fee Treasury; swept weekly to cold storage.				
BANKROLL ALLOCATION BUCKET	SATOSHI AMOUNT	BSV AMOUNT	USD EQ. (\$14/BSV)	OPERATIONAL PURPOSE & GUARDRAIL
LMSR Market Liquidity Reserve (Bootstrap Phase)	1,386,300 sats	0.0138 BSV	\$0.19	Covers exact maximum mathematical loss for 10 initial active markets (138,630 sats each).
Hot Wallet Operating Balance	100,000,000 sats	1.0000 BSV	\$14.00	Funds on-chain transaction broadcast fees and immediate withdrawal sweeps.
Customer Withdrawal Bootstrap Buffer	500,000,000 sats	5.0000 BSV	\$70.00	Initial treasury float ensuring instant 0-conf withdrawals during launch ramp-up.
Emergency Liquidity Buffer	398,613,700 sats	3.9861 BSV	\$55.81	Contingency fund allowing immediate expansion up to 20 markets without capital injection.
TOTAL RECOMMENDED STARTING BANKROLL (STAGE 1)	1,000,000,000 sats	10.0000 BSV	\$140.00	Exact recommended bootstrap testing bankroll (1 BSV = 100,000,000 satoshis).

CONCRETE POOL PROTECTION CONTROLS

1. The system checks free bankroll before transitioning any market from `Reviewed` to `Open`. If free reserve is below `138,630 satoshis`, the market remains locked.
2. The system enforces strict risk caps during Stage 1: `10-20` maximum active markets, `1,000,000 satoshis` maximum order size, and `5,000,000 satoshis` maximum position limit per user per market.
3. If internal double-entry reconciliation detects any mismatch ($\text{Total User Balances} \neq \text{Hot Wallet Deposit Total}$), all trading halts automatically within `100ms`.

Recommended starting bankroll: 10 BSV / 1,000,000,000 satoshis (\$140 USD equivalent at \$14/BSV assumption on July 2026).

9.1 SOLVING LOW INITIAL PAYOUTS: THE 3-STAGE DYNAMIC LIQUIDITY & PAYOUT SCALING STRATEGY

The Architectural Challenge of a 10 BSV Pool: Why does the initial Phase 1 MVP limit single orders to `1,000,000 satoshis` (`\$0.14`) and keep the `b` parameter at `200,000 satoshis`? Because in an LMSR engine, the maximum potential loss per market is directly proportional to `b` ($b * \ln(2)$). If the platform opened right on Day 1 with `b = 100,000,000 satoshis` (`1 BSV`), a single unexpected resolution discrepancy across 10 markets could theoretically drain `10 BSV`. By starting with `b = 200,000 satoshis`, the platform's total worst-case loss across 10 active markets is strictly locked at **1,386,300 satoshis` (`0.0138 BSV` / `\$0.19`)**, protecting over **99% of the 10 BSV bankroll** while real users live-test 0-conf deposits, withdrawals, and Kalshi resolution.

How We Scale Payouts Without Compromising Pool Safety: As soon as the platform completes its verification phase and organic trading volume flows in, the BSVA executes a predetermined ****3-Stage Dynamic Expansion Model****. By scaling `b` from `200,000 sats` up to `5,000,000 sats` and ultimately `100,000,000 sats`, price slippage drops near zero, allowing traders to execute ****\$3.50, \$70.00, and \$280.00+ single orders**** while keeping the liquidity pool 100% mathematically protected by the expanded bankroll.

DIAGRAM 6A — THE 3-STAGE DYNAMIC LIQUIDITY & PAYOUT SCALING MATRIX

Stage 1: Public Bootstrap

10 BSV Pool (\$140)

Goal: Live production testing, zero-risk verification of double-entry ledger, and initial community onboarding across 10 curated markets.

LMSR Parameter ('b'):	200,000 sats (0.002 BSV)
Max Single Trade:	1,000,000 sats (\$0.14)
Max User Position:	5,000,000 sats (\$0.70)
Max Loss / Market:	138,630 sats (\$0.019)
Slippage Profile:	High (Ideal for micro-bets)

Stage 2: Growth Scaling

100 BSV Pool (\$1,400)

Trigger: After 30 days of zero accounting errors and daily volume exceeding '10 BSV'. BSVA injects '100 BSV' ('\$1,400') and scales 'b' by 25x.

LMSR Parameter ('b'):	5,000,000 sats (0.05 BSV)
Max Single Trade:	25,000,000 sats (\$3.50)
Max User Position:	100,000,000 sats (\$14.00)
Max Loss / Market:	3,465,735 sats (\$0.48)
Slippage Profile:	Tight (Supports \$1-\$10 bets)

Stage 3: Enterprise Maturity

1,000 BSV Pool (\$14,000)

Trigger: High-demand major world events (e.g., US Elections or Bitcoin halving) with proven high-throughput 24/7 ledger resilience.

LMSR Parameter ('b'):	100,000,000 sats (1.0 BSV)
Max Single Trade:	500,000,000 sats (\$70.00)
Max User Position:	2,000,000,000 sats (\$280.00)
Max Loss / Market:	69,314,718 sats (\$9.70)
Slippage Profile:	Deep Liquidity (Whale ready)

WHY ORGANIC TRADING FEES ('1.0%') ELIMINATE POOL SUBSIDY LOSS OVER TIME

An LMSR market maker only absorbs its maximum $b \cdot \ln(2)$ loss if trading is entirely one-sided. Every time users trade back and forth (buying both YES and NO as market news evolves), the platform collects a **1.0% flat fee on trade volume**. Once a market attracts balanced two-way volume exceeding $100 \cdot b$, the accumulated 1% fee revenue inside the Fee Treasury fully pays for the initial liquidity subsidy ($b \cdot \ln(2)$), turning the market maker **net-profitable** while continuously offering deep, instant order matching!

10. FOUR-WEEK INTENSIVE EXECUTION ROADMAP

A disciplined, fast-track 4-week engineering schedule executed entirely by internal BSV Association (BSVA) team members (\$0 external labor cost).

DIAGRAM 7 — FOUR-WEEK DEVELOPMENT TIMELINE ACROSS 4 SPRINT PHASES

Phase 1: Product, Specs & Ledger	W1			
Phase 2: Wallet, Deposits & Kalshi Sync		W2		
Phase 3: Trading Engine & Settlement			W3	
Phase 4: Audit, QA & Public Launch				W4

Phase 1 (Week 1) — Product Definition, UI Specification & Core Double-Entry Ledger

DURATION: WEEK 1

Cost: \$0 Labor (Internal BSVA)

SPECIFIC ACTIONS

- Freeze exact MVP scope, user journey, and binary trading rules.
- Specify LMSR exact parameters (`b = 200,000 sats`, 1% fee, `100,000 sat` contract standard).
- Design clean UI wireframes for Catalogue, Market Detail, Portfolio, and Admin screens.
- Construct off-chain double-entry database & accounting schema in PostgreSQL and Redis on staging VPS.

DELIVERABLE OUTPUT

- Approved Product Specification Document.
- Figma UI Screen Designs.
- Live Database & Double-Entry Ledger Schema.

TEAM & EFFORT

- **People:** BSVA Product Lead, UI Designer, Lead Senior Backend Dev.
- **Work:** 1-Week Intensive Sprint (Internal Staff).

Phase 2 (Week 2) — BSV Wallet Connection (BEEF/SPV), Deposit Detection & Kalshi Provider Adapter

DURATION: WEEK 2

Cost: \$0 Labor (Internal BSVA)

SPECIFIC ACTIONS

- Implement browser wallet login with standard BEEF/SPV support for Sensilet, Panda, or Yours.
- Build hot wallet deposit monitoring worker generating unique derivation references per user.
- Build modular Kalshi REST/Webhook adapter to ingest binary questions, schedules, and resolution criteria.
- Construct internal admin review dashboard to curate and approve imported markets.

DELIVERABLE OUTPUT

- Working on-chain deposit detection & crediting.
- Live Kalshi market synchronization.
- Curated market catalogue & admin interface.

TEAM & EFFORT

- **People:** 2 BSVA Full-Stack Devs, BSV Integration Specialist.
- **Work:** 1-Week Intensive Sprint (Internal Staff).

Phase 3 (Week 3) — Off-Chain LMSR Trading Engine & Automated Outcome Settlement

DURATION: WEEK 3

Cost: \$0 Labor (Internal BSVA)

SPECIFIC ACTIONS

- Implement integer-safe LMSR pricing algorithm in Redis/TypeScript (10^8 scaling factor).
- Build instant buy/sell quotation endpoints computing exact price impact, fee, and share issuance.
- Enforce exposure guardrails (5,000,000 sats max per user) and reserve 138,630 sats bankroll loss per market.
- Build automated market closing scheduler and outcome resolution settlement worker (100,000 sats winner crediting).

DELIVERABLE OUTPUT

- High-speed buy/sell execution engine.
- Real-time dynamic YES/NO price discovery.
- Automated end-to-end outcome settlement flow.

TEAM & EFFORT

- **People:** 2 BSVA Full-Stack Devs, Lead Senior Backend Dev.
- **Work:** 1-Week Intensive Sprint (Internal Staff).

Phase 4 (Week 4) — Rigorous Testing, Audit Verification & Public MVP Launch

DURATION: WEEK 4

Cost: \$0 Labor (Internal BSVA)

SPECIFIC ACTIONS

- Execute automated stress testing across 100,000 simulated trades to verify exact integer accounting.
- Deploy automated hot wallet satoshi withdrawal broadcast service and emergency admin pause switches.
- Fund production hot wallet (1.0 BSV) and reserve 1,386,300 sats for 10 initial curated markets.
- Publish the 10 curated binary prediction markets to the production web application on dedicated VPS servers!

DELIVERABLE OUTPUT

- Verified zero-error accounting audit report.
- Live, public BSV Prediction Market MVP.
- Operational monitoring & telemetry dashboards.

TEAM & EFFORT

- **People:** Full BSVA Team + QA Security Specialist.
- **Work:** 1-Week Intensive Sprint (Internal Staff).

11. RECOMMENDED IMPLEMENTATION TEAM (INTERNAL BSVA STAFF)

Because this MVP will be developed and maintained entirely by internal **BSV Association (BSVA) employees**, there is **\$0 external contractor cost** for salaries or development labor. Build and launch this MVP with the following 5 internal roles across the **4-week intensive sprint** to maintain maximum execution velocity without management overhead:

INTERNAL ROLE & TITLE	ALLOCATION (4-WEEK SPRINT)	PRIMARY RESPONSIBILITIES ACROSS PHASES
BSVA Product Lead / Architect	Full-Time (4 weeks)	Owns product specification, Kalshi market curation, trading rules, risk limits, and admin operations.
BSVA Senior Full-Stack Engineer (Lead Backend)	Full-Time (4 weeks)	Builds double-entry accounting ledger, integer LMSR engine, Redis state locking, and settlement worker.
BSVA Full-Stack Engineer (Frontend & API)	Full-Time (4 weeks)	Develops TypeScript/React web app, trading UI, portfolio view, Kalshi REST adapter, and admin dashboard.
BSVA BSV Integration Specialist	Part-Time (1.5 weeks equivalent)	Implements BEEF/SPV wallet connection, on-chain deposit monitoring worker, and withdrawal signing service.
BSVA QA & Security Test Engineer	Part-Time (1.5 weeks equivalent)	Conducts rigorous integer accounting audits, stress testing, double-spend checks, and pre-launch validation.

12. REALISTIC MVP DEVELOPMENT BUDGET (STAGING VPS & TOOLING — 4-WEEK BUILD)

Because all engineering labor is provided by existing internal BSVA staff (`\$0 personnel cost`), the MVP development budget consists of practical staging VPS instances, managed database hosting, commercial API access, and basic monitoring during the 4-week build.

INFRASTRUCTURE / TOOLING BUDGET ITEM (4-WEEK BUILD)	PROVIDER & SPECIFICATION DETAILS (STAGING VPS SERVERS & API SUBSCRIPTIONS)	ASSIGNED INTERNAL PERSONNEL	COST (USD)
1. Internal BSVA Engineering Labor (5 Roles across 4 Weeks)	1 Product Lead, 2 Full-Stack Devs, 1 BSV Specialist, 1 QA Engineer	Internal BSVA Staff	\$0
2. Staging & Dev VPS Servers (1 Month Setup)	`3x` Dedicated VPS instances on Hetzner/IONOS for staging API gateway, database testing, and workers	BSVA Senior Backend Dev	\$280
3. Managed Staging Database & Redis Cache	Managed PostgreSQL + Redis instance with SSD storage and automated testing snapshots for dev environment	BSVA Senior Backend Dev	\$180
4. Kalshi Commercial API Developer Subscription	Commercial API tier for market metadata ingestion, sandbox testing & live webhook resolution verification	BSVA Frontend/API Dev	\$450
5. BSV SPV & Wallet Verification Dev Server	Dedicated VPS instance (`64GB RAM tier`) hosting SPV deposit listeners & BEEF transaction verification workers	BSVA Integration Specialist	\$140
6. Domain, Cloudflare DNS & Sentry Error Tracking	Cloudflare Pro DNS/WAF protection (`\$25`), domain setup, and Sentry APM error logging team plan (`\$80`)	BSVA Product Lead	\$150
TOTAL MVP DEVELOPMENT INFRASTRUCTURE BUDGET (4 WEEKS)			\$1,200 USD

13. MONTHLY OPERATING COST & TOTAL INITIAL CASH REQUIREMENT

Ongoing monthly expenditure after public launch running on reliable dedicated VPS servers with internal BSVA maintenance (\$0 external labor).

MONTHLY OPERATING EXPENSE CATEGORY	INFRASTRUCTURE / PROVIDER DETAILS (PRODUCTION DEDICATED VPS SERVERS)	MONTHLY COST (USD)
Internal BSVA Operational & Admin Maintenance	Managed directly by internal BSVA team members during regular operations	\$0
Production Application VPS Servers	`4x` Dedicated VPS servers on Hetzner/IONOS for API gateway, web application, and background workers	\$480
Managed Production Database & Redis	Managed PostgreSQL database instance (`2x` redundant nodes with automated daily backups) + Redis cache	\$350
BSV Blockchain SPV Worker Servers	`2x` Dedicated VPS instances running SPV deposit listeners, BEEF verification, and broadcast workers	\$220
Kalshi Commercial API Production Feed	Commercial API data subscription & real-time resolution webhook pipeline with guaranteed SLA	\$450
DevOps, Domain, SSL & Sentry Error Monitoring	Cloudflare DNS/WAF shield, Sentry error logging, automated backup snapshot storage, and uptime checks	\$200
TOTAL MONTHLY OPERATING COST (POST-LAUNCH)		\$1,700 USD / mo

TOTAL INITIAL OUT-OF-POCKET CASH REQUIREMENT SUMMARY (\$14/BSV ASSUMPTION)

4-Week Development Infrastructure Budget: \$1,200 USD (`\$0` labor — built by internal BSVA staff on staging VPS servers)

Starting Bankroll Liquidity Capital (Stage 1 Bootstrap): \$140 USD (`10.0 BSV` / `1,000,000,000 satoshis` at \$14/BSV)

First Month Production Operating Expense: \$1,700 USD (`Production VPS servers, managed database & API feeds`)

Total Initial Out-of-Pocket Cash Requirement: \$3,040 USD

14. EXACT MVP SCOPE CHECKLIST & EXCLUDED FEATURES (“LATER”)

Strict feature boundaries enforcing immediate launch velocity and preventing scope creep.

Included in Public MVP Scope	WEEK 4 DELIVERABLES	Excluded Features	DEFERRED TO LATER
☒ 10–20 Curated Binary Markets ☒ BSV Wallet Connection (SPV / BEEF) ☒ Internal Satoshi Balance Tracking ☒ Dynamic Off-Chain LMSR Prices ☒ User Portfolio & Transaction History ☒ Winner Balance Credit (`100,000 sats`) ☒ Admin Review, Pause & Refund Controls	☒ Kalshi External Provider Adapter ☒ 0-Conf Satoshi Deposit Detection ☒ Market Catalogue & Detail Pages ☒ Instant Buy & Sell Position Execution ☒ Automated Market Close & Settlement ☒ Automated Satoshi Withdrawal Engine ☒ Exposure Guardrails (`5M sats` limit)	☒ User-Created Markets ☒ Order Books & Limit Orders ☒ Platform Utility or Governance Tokens ☒ On-Chain Smart Contract AMM State ☒ Social Feeds & Copy Trading ☒ Multiple Simultaneous Result Providers	☒ Multi-Outcome / Categorical Markets ☒ Leverage, Margin & Loans ☒ External Liquidity Provider (LP) Pools ☒ Native iOS & Android Mobile Apps ☒ Advanced Multi-Tier Referral Systems ☒ Hundreds of Simultaneous Active Markets

15. IMMEDIATE ACTION PLAN (WEEK 1, WEEK 2, WEEK 3, WEEK 4, AND POST-LAUNCH GROWTH)

Concrete, time-bound execution milestones starting from Day 1 to achieve full public launch in 30 days.

Week 1 (Days 1–7) — Foundation, Spec & Schema

- **Freeze the MVP scope:** Sign off on the exact inclusion/exclusion list.
- **Define contract standard:** Set fixed `100,000 satoshis` (\$0.014) payout unit.
- **Set LMSR parameter:** Lock `b = 200,000 satoshis` and `1.0%` trading fee for Stage 1.
- **Define risk limits:** Set `1M sats` order cap and `5M sats` user position cap.
- **Create database schema:** Finalize double-entry PostgreSQL and Redis data structures.
- **Design core screens:** Complete wireframes for Catalogue and Trading UI.

Week 2 (Days 8–14) — Wallet Login, Deposits & Kalshi Feeds

- **Complete wallet connection:** Deploy working browser wallet login (BEEF/SPV).
- **Complete deposit detection:** Verify 0-conf satoshi hot wallet crediting.
- **Complete internal ledger:** Enforce strict double-entry accounting invariants.
- **Import first markets:** Synchronize 10 binary questions from Kalshi API adapter.
- **Display opening quotes:** Render real-time `50,000 sat` opening quotes in UI.

Week 3 (Days 15–21) — Trading Engine & Settlement Flow

- **Complete buy & sell flows:** Verify instant position issuance and closing quotes.
- **Verify integer scaling:** Confirm zero rounding discrepancy across `10^8` LMSR math.
- **Complete settlement:** Automate Kalshi outcome import and winner balance crediting (`100,000 sats`).
- **Complete admin dashboard:** Enable emergency market pause, void, and refund overrides.

Week 4 (Days 22–28) — QA Audit & Public MVP Launch

- **Execute automated audit:** Verify 100,000 simulated trades with zero accounting error.
- **Deploy withdrawal worker:** Activate automated hot wallet satoshi broadcast service.
- **Fund liquidity reserve:** Deposit exactly `10 BSV` (`1,000,000,000 sats`) into bankroll for Stage 1.
- **Launch public MVP:** Open real satoshi deposits and publish 10 curated markets on dedicated production VPS servers!
- **Trigger Stage 2 scaling:** Inject `100 BSV` and scale `b` by `25x` once post-launch organic targets are met.

16. FINAL PRE-LAUNCH VERIFICATION CHECKLIST

Do not open public access until every item below is checked and verified in production during Week 4.

✓ Production hot wallet balance confirmed at exactly `1.0 BSV` (`100,000,000 satoshis`).

✓ Market liquidity bankroll confirmed at exactly `1,386,300 satoshis` locked in reserve for Stage 1.

✓ Customer withdrawal reserve pool confirmed separated and backed 100% by deposited funds.

✓ Kalshi provider adapter verified receiving real-time webhook status updates.

✓ Double-entry accounting invariant (`Sum of Balances == Net Deposits`) verified via cron alert.

✓ LMSR integer scaling (`10^8` factor) verified across `1,000 sats` to `1,000,000 sats` orders.

✓ Emergency admin pause switch tested and confirmed halting orders within `100ms`.

✓ Automated daily database backups and Redis snapshot persistence active on production VPS servers.